

Xinkai Chen

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Machine Learning Engineer with **5+ years of experience** building and shipping production ML systems — spanning LLMs, Transformers-based NLP, and gradient boosted trees — driving **millions** in annual savings and earning a US patent.

EXPERIENCE

Data Scientist

Elevance Health, Inc. (formerly Anthem Inc.)

Chicago, IL

Mar 2021 – Present

Built and owned production ML and LLM systems to automate clinical authorization decisions across 30+ medical guidelines and multiple business units, saving tens of thousands of physician and nurse hours annually.

- Built a production-grade **VLM** pipeline to extract behavioral assessment scores from **10K+** charts at **99%** accuracy, reducing latency by **90%** while serving as core quality metrics for downstream anomaly detection tasks
- Core contributor to the MLOps ecosystem — built pip-installable packages using **UV**, **Poetry**, and **JFrog**, and integrated **GitLab CI/CD** pipelines for linting, testing and deployment, reducing release cycle time by **50%**
- Designed a **DAG-based LLM** pipeline for therapy visit recommendations with swappable nodes, Pydantic-enforced outputs, and robust retry logic, achieving **90%** accuracy and **~\$200K/month** in savings
- Implemented hybrid **RAG** with section-aware chunking, BM25 sparse and OpenAI embedding dense retrieval via Qdrant, and cross-encoder reranking to enhance search quality, reducing context token usage by **40%**
- Trained **LightGBM** models across 15+ medical guidelines, tuned decision thresholds for target FP rates and applied SHAP for clinical explainability — enabling **30%** automatic case approvals and **~\$100K** monthly savings
- Built clinical NLP pipelines using **UmlsBERT/CODER** embeddings and **spaCy/Stanza** to extract structured medical entities from unstructured clinical notes, powering downstream authorization models at **85% precision**
- **Led a pod of 6 data scientists**, building a scalable tree-based codebase that could be rapidly adapted across clinical guidelines — driving **25+** model deployments in five months with minimal per-use-case overhead
- **Patented** a novel system for authorization automation using Artificial Intelligence ([US 12293835 B1](#))

Machine Learning Consultant

Hamiltonian Systems, Inc.

Pittsburgh, PA

Jun 2020 – Dec 2020

- Designed a multithreaded **Flask API** to orchestrate ML model training and inference pipelines with async task management and cloud deployment via REST endpoints — reducing manual engineering workload by **90%**
- **Led a team of 5** to build inventory lifecycle prediction models using **Random Forest** and **XGBoost** on 440K+ transactions cleaned and transformed with Oracle SQL and Pandas, reducing forecast error by 22% over baseline

PROJECTS

Cerberus: Real-Time Fraud Detection ML Pipeline | *Kafka, Feast, Redis, FastAPI, Docker* Jan – Mar 2026

- Developed a feature engineering pipeline using **Kafka** and **Feast**, synchronizing real-time data to **Redis** and historical records to PostgreSQL to achieve **sub-10ms** feature retrieval latency
- Integrated an XGBoost model to identify high-risk transactions and containerized the entire lifecycle (producer, stream processor, and FastAPI inference service) using **Docker**

MacroForge: Multi-Agent Nutrition System | *Google ADK, Gemini, GCP, Vertex AI* Oct – Nov 2025

- Architected a multi-agent system using Google **Agent Development Kit** (ADK) to autonomously coordinate **specialized sub-agents** for recipe generation, food substitution, and shopping optimization
- Orchestrated a resilient, grounded agent pipeline by integrating **custom tools**, MCP servers, and Google Search APIs, with dynamic routing and **Vertex AI** for scalable production deployment

EDUCATION

Carnegie Mellon University

Master of Information Systems Management (GPA: 3.91, Highest Distinction)

Pittsburgh, PA

Aug 2019 – Dec 2020

Fudan University

Bachelor of Information Security (School of Computer Science)

Shanghai, China

Sep 2015 – Jun 2019

TECHNICAL SKILLS

ML Frameworks: Pytorch, TensorFlow, Transformers, Hugging Face, LightGBM, XGBoost, Scikit-learn, spaCy

MLOps: AWS (Kubeflow, SageMaker), GCP (Vertex AI), Docker, FastAPI, GitLab CI/CD, JFrog, Poetry, UV

Data & Streaming: Kafka, Redis, PostgreSQL, MongoDB, Snowflake, Feast, Spark

Large Language Models: OpenAI SDK, Claude Code CLI, LangChain, Google ADK, Pydantic, DSPy, Qdrant